

Mandatory assignment 1

Deadline: Saturday September 27th at 23:59 (Norwegian time).

Read carefully through the information about the mandatory assignments on Canvas. Notice in particular that the assignments should be solved individually.

Hand in on Canvas. Submissions should be of **either** of the following types

- Submit two files: One pdf-file with a report containing the answers to the theory questions, and one R-file including the R-code.
- Submit two files: One R markdown (Rmd) file containing both theory answers and R-code, and a pdf-file with the output you obtain when running (knitting) your R markdown file. See tutorial to get started.

The first line of R-code should be: `rm(list=ls())`. Check that the Rmd/R-code file runs before you submit it. Use comments in the R-code to clearly identify which question each part of the R-code belongs to. Also try to add some comments to explain important parts of the code. The file ending of the R-code file should be .Rmd, .R or .r. The report can be handwritten and scanned to pdf-file, or written in your choice of text editor and converted to pdf. Cite the sources you use.

Problems marked with an ^R should be solved in R, the others are theory questions. Each of the subproblems, e.g., points 1a), 1b), 2a), and so on, is given the same weight.

Problem 1:

Meteorologists in Stavanger are studying daily maximum wind speeds (in m/s) during autumn. Based on historical records, it is believed that the daily maximum wind speed X on a randomly chosen autumn day never falls below a minimum threshold $a > 0$ m/s and never exceeds $a + 20$ m/s. To model this behaviour, the following probability density function is proposed:

$$f(x; a) = \begin{cases} c(x - a)(a + 20 - x) & , \quad a \leq x \leq a + 20 \\ 0 & , \quad \text{otherwise} \end{cases}$$

where $c > 0$ is a normalising constant.

- Show that $f(x; a)$ is a valid pdf by finding the constant c (expressed in terms of a). *Hint: Use the substitution $u = x - a$.*
- Compute the expected daily maximum wind speed $E(X)$ and the variance $\text{Var}(X)$, both expressed as functions of a . Interpret your results in the context of wind speed in Stavanger.

- c) Show that the cumulative distribution function $F(x; a)$ for $a \leq x \leq a + 20$, expressed as a function of a , is given by

$$F(x; a) = \frac{3}{4000} \left(10(x - a)^2 - \frac{(x - a)^3}{3} \right), \quad a \leq x \leq a + 20.$$

- d)^R Recall from the lectures that if $U \sim U(0, 1)$, then $X = F^{-1}(U)$. To apply this, we need to solve $F(X; a) = U$ for X . Since $F(x; a)$ is a cubic polynomial in $(x - a)$, inverting it analytically is not straightforward and leads to a complex expression involving trigonometric functions. Instead, we can solve $F(X; a) = U$ numerically in R using `uniroot()`: for each realisation u of U , the call `uniroot(function(x) F(x,a) - u, lower=a, upper=a+20)` finds the unique X such that $F(X; a) = u$, i.e. $X = F^{-1}(U; a)$.

Using this approach, write an R function that implements $F^{-1}(p; a)$ via `uniroot`, and use it to generate $n = 5000$ random wind speed samples with $a = 5$. Then:

- i) Plot a histogram of the simulated values using the Scott rule and overlay the true density $f(x; a = 5)$. Comment on the fit.
 - ii) Plot the empirical CDF (ECDF) and overlay the true CDF $F(x; a = 5)$. Comment on the fit.
- e) As an alternative to the inverse transform method, we can use the Acceptance-Rejection (AR) method to simulate from $f(x; a)$. We use a uniform proposal distribution $g(x)$ on $[a, a + 20]$.
- i) Find the constant M such that $f(x; a) \leq M \cdot g(x)$ for all x .
 - ii) Describe the AR algorithm step by step for this specific choice of f and g .
 - iii) What is the theoretical acceptance rate of the algorithm? How many proposals do you expect to need in order to generate $n = 5000$ accepted samples?
- f)^R Implement the Acceptance-Rejection algorithm from e) in R with $a = 5$ and generate $n = 5000$ samples. Then:
- i) Plot a histogram of the simulated values and overlay the true density $f(x; a = 5)$. Comment on the fit.
 - ii) Report the empirical acceptance rate from your simulation and compare it to the theoretical value from e)iii).
 - iii) Compare briefly the two simulation approaches from d) and f): when would you prefer one over the other?

Problem 2:

A coastal weather station near Stavanger records daily rainfall (in mm) over a long period. Rainfall on a wet day is modelled by an Exponential distribution with scale parameter $\beta > 0$, i.e. with pdf:

$$f(x; \beta) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0.$$

A sample of $n = 40$ wet-day rainfall measurements (in mm) is given below:

```
rain <- c(3.1, 7.2, 5.4, 12.0, 8.8, 4.3, 9.6, 6.1, 11.2, 2.9,  
          7.5, 14.1, 5.0, 9.3, 6.8, 10.4, 3.7, 8.1, 13.5, 4.6,  
          6.3, 11.8, 7.9, 5.5, 9.0, 12.7, 4.1, 8.4, 6.6, 10.1,  
          3.4, 7.0, 5.8, 13.2, 9.7, 6.2, 11.5, 4.8, 8.9, 7.3)
```

- a) Based on $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\beta)$, show that the Maximum Likelihood Estimator (MLE) of β is $\hat{\beta} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.
- b)
- i) Is $\hat{\beta}$ an unbiased estimator of β ? Justify your answer.
 - ii) Show that $\text{Var}(\hat{\beta}) = \beta^2/n$.
- c) How many wet-day rainfall measurements would we need to collect to be 95% confident that $\hat{\beta}$ falls within 1 mm of the true β ? Justify your answer using the CLT.
- d)^R Using the data vector `rain` given above:
- i) Compute the MLE $\hat{\beta}$ and report it.
 - ii) Construct an approximate 95% confidence interval for β using the CLT. Report and interpret the interval.

Problem 3:

We continue with the rainfall data from Problem 2 and study the distribution of wet-day rainfall more carefully using graphical tools.

- a)^R Plot the ECDF of the `rain` data. On the same plot, overlay the theoretical CDF of an $\text{Exp}(\hat{\beta})$ distribution using your MLE from Problem 2d).
- b)^R Produce a histogram of the `rain` data (using Scott's rule for bin width). On the same plot, overlay:
 - i) The fitted Exponential pdf with parameter $\hat{\beta}$.
 - ii) A Kernel Density Estimate (KDE) using R's `density()` function with the default bandwidth.
- c) Based on your observations in a) and b):
 - i) Does the Exponential distribution provide a good fit to the rainfall data? Argue and explain.
 - ii) Can you think of an alternative distribution that might provide a better fit? Briefly justify your choice.
- d)^R Estimate the parameters of your chosen distribution from c)ii) using `fitdistr()` from the `MASS` package in R. This function finds the MLEs numerically — it can be called as `fitdistr(rain, "name_of_distribution")`. Produce a plot with the histogram of the `rain` data and overlay:
 - i) The fitted density from your alternative model.
 - ii) KDE curves using three bandwidths: one clearly too small, one default (via `bw.nrd0()`), and one clearly too large.

Comment on how the alternative parametric model compares to the KDE across the three bandwidths, and what this tells you about the bias-variance tradeoff in bandwidth selection.

Problem 4:

The total annual precipitation at the Stavanger weather station is modelled as the sum of contributions from four independent sources: orographic rainfall A , frontal rainfall B , convective rainfall C , and snowfall D . Their distributions are:

Source	Random variable	Distribution
Orographic	A	Normal with $\mu = 400$, $\sigma^2 = 900$
Frontal	B	Exponential with mean 250
Convective	C	$U(50, 150)$
Snowfall	D	80 with probability 0.3, 0 with probability 0.7

All values are in mm. The total annual precipitation is $Y = A + B + C + D$. The four sources are assumed to be independent.

- a) Find the expected total annual precipitation $E(Y)$ and the standard deviation $SD(Y)$.
- b)^R Estimate the distribution of Y by simulation using $n = 5000$ samples.
 - i) Draw a histogram of the simulated values of Y .
 - ii) What are the estimated mean and standard deviation of Y ? Compare with your analytical results from a).
 - iii) Estimate the probability $P(Y > 1050)$ by simulation and interpret the result.